

## **EXPERIMENT 2**

### **Problem Identification and Feasibility: Software Design Principles**

#### **Selected Project: Online Mutual Fund Portal**

##### **Aim**

To identify the problem addressed by the **Online Mutual Fund Portal** and prepare its problem statement, objectives, scope, and technical and operational feasibility analysis.

##### **Objectives**

- Identify the real-world problem addressed by the Online Mutual Fund Portal.
- Prepare a clear problem statement for digitizing mutual fund discovery, comparison, investment and transaction management.
- Define the objectives and scope of the proposed web-based mutual fund investment solution.
- Perform technical and operational feasibility analysis using the proposed software architecture.
- Evaluate the viability of the proposed solution for investors and administrators.

##### **Introduction**

Traditional mutual fund investment processes may require investors to depend on intermediaries or visit financial service offices to obtain information about schemes, compare funds and manage investments. Investors may also face difficulties in tracking their portfolios, reviewing transaction history and monitoring investment performance from a single platform.

The **Online Mutual Fund Portal** digitizes these activities into a centralized web-based system. It provides authenticated access for users and administrators, allowing investors to browse and search mutual fund schemes, view fund details, compare different schemes, invest in mutual funds, manage their portfolios and track transaction history. Administrators can manage mutual fund schemes, update fund information and monitor user investments and transactions.

##### **Selected Project**

**Project Title:** Online Mutual Fund Portal

**Domain:** Financial Technology / Investment Management

**Platform:** Web-based application

##### **Problem Statement**

Investors often face difficulties in finding relevant mutual fund schemes, comparing fund information and managing their investments through a single platform. Manual or fragmented processes can make it inconvenient to track investment portfolios, transaction history and investment performance.

The **Online Mutual Fund Portal** provides a centralized digital platform where users can register and securely log in, browse and search mutual fund schemes, view detailed fund information, compare different schemes, invest in mutual funds and manage their investment portfolios. The system also maintains transaction and investment history for users. Administrators can manage mutual fund schemes, update fund information, manage user accounts and monitor investments and transactions.

### **Project Objectives**

- Provide secure user registration and login functionality.
- Allow users to browse and search available mutual fund schemes.
- Provide detailed information about mutual fund schemes.
- Allow users to compare multiple mutual fund schemes.
- Enable users to invest in selected mutual fund schemes.
- Provide users with an investment portfolio for tracking their investments.
- Maintain and display transaction and investment history.
- Allow users to track the performance of their investments.
- Provide administrators with facilities to add, update and remove mutual fund schemes.
- Allow administrators to manage fund information, user accounts and system transactions.
- Maintain secure and role-based access for users and administrators.

### **Project Scope**

The Online Mutual Fund Portal covers user registration and authentication, mutual fund scheme browsing, searching and filtering, fund-detail viewing, fund comparison, investment management, portfolio tracking and transaction-history management.

The system also provides an administrator portal through which administrators can add, update and remove mutual fund schemes, manage fund information, manage user accounts and monitor investments and transactions.

The proposed system uses **HTML, CSS and JavaScript** for the frontend, **Java/Python/Node.js** for backend services and **MySQL** for relational data storage. Git

and GitHub can be used for source-code management, collaboration and version control.

## **Major Modules**

### **1. User Authentication Module**

- User registration
- User login
- Authentication and authorization
- Role-based access

### **2. Mutual Fund Management Module**

- Browse funds
- Search and filter funds
- View fund details
- Compare mutual funds

### **3. Investment Module**

- Select mutual fund schemes
- Make investment requests
- Maintain investment records
- Track investment status

### **4. Portfolio Management Module**

- View current investments
- Track investment performance
- View investment history

### **5. Transaction Management Module**

- Record investment transactions
- View transaction history
- Monitor transaction information

### **6. Administrator Module**

- Admin authentication

- Add, update and remove mutual fund schemes
- Manage fund information
- Manage user accounts
- Monitor investments and transactions

### Technical Feasibility Analysis

| Factor                | Assessment | Remarks                                                                                                       |
|-----------------------|------------|---------------------------------------------------------------------------------------------------------------|
| Technology            | Feasible   | HTML, CSS, JavaScript, Java/Python/Node.js and MySQL support development of the system                        |
| Web Infrastructure    | Feasible   | Standard web-server and database infrastructure can support the portal                                        |
| Authentication        | Feasible   | User and administrator authentication can be implemented using backend authentication mechanisms              |
| Database              | Feasible   | MySQL can store users, mutual funds, investments, portfolios and transaction records                          |
| Investment Management | Feasible   | Backend services can process and maintain investment-related records                                          |
| Search and Comparison | Feasible   | Database queries can support fund searching, filtering and comparison                                         |
| Scalability           | Feasible   | Modular backend architecture can support future expansion of users and mutual fund schemes                    |
| Version Control       | Feasible   | Git and GitHub support collaborative development and source-code management                                   |
| Security              | Feasible   | Authentication, authorization, input validation and database security mechanisms can protect application data |

The technical feasibility assessment indicates that the proposed Online Mutual Fund Portal can be implemented using the selected web technologies and database system. The application can be divided into independent modules such as authentication, fund management, investment management, portfolio management and administration.

MySQL provides structured storage for user accounts, mutual fund schemes, investments and transaction records. Backend technologies such as Java, Python or Node.js can implement business logic and API services, while HTML, CSS and JavaScript can provide the user interface.

### Operational Feasibility Analysis

| Factor                    | Assessment | Remarks                                                                                 |
|---------------------------|------------|-----------------------------------------------------------------------------------------|
| User Acceptance           | High       | Users can access mutual fund information and manage investments through a single portal |
| Ease of Use               | High       | Search, filtering, comparison and portfolio features simplify investment management     |
| Accessibility             | High       | The web-based system can be accessed through standard web browsers                      |
| Maintainability           | High       | Modular separation of system components makes maintenance easier                        |
| Security                  | High       | Authentication and role-based authorization separate user and administrator operations  |
| Administrative Management | High       | Administrators can centrally manage schemes, users and transaction information          |
| Operational Benefit       | High       | Centralized fund information and investment management reduce fragmented processes      |

### Analysis

The feasibility study indicates that the **Online Mutual Fund Portal is technically and operationally viable** based on the proposed architecture and technologies.

The selected technologies provide the necessary components for developing the portal. HTML, CSS and JavaScript can be used to create an interactive frontend, while Java, Python or Node.js can provide backend services. MySQL provides structured storage for mutual fund, user, investment and transaction data.

The system can reduce the need for fragmented investment management by providing fund discovery, comparison, investment management and portfolio tracking through a single platform. The administrator module also provides centralized management of mutual fund schemes, users and transactions.

The modular architecture allows future features to be added, such as online payment integration, notifications, advanced portfolio analytics, SIP management and integration with external financial data services.

### **Observation**

Clearly defining the investment workflow, user and administrator roles, system modules, scope and feasibility provides a strong foundation for the subsequent software design and development of the **Online Mutual Fund Portal**.

The feasibility analysis demonstrates that the proposed system can be developed using commonly available web technologies and a relational database while providing centralized investment-management functionality.

### **Result**

The problem statement, objectives, project scope, and technical and operational feasibility analysis for the **Online Mutual Fund Portal** were successfully prepared. The study establishes a feasible foundation for the design and development of a web-based platform for mutual fund discovery, comparison, investment and portfolio management.